causes a similar change in the collector current. Now, the collector current flows through a high load resistance R_L, which produces a large voltage across R_L.

Thus, the weak signal applied in the input circuit appears in the amplified form in the collector circuit. In this way, transistor acts as an amplifier. For power amplification, heat sinks are used to fix the transistor in order to dissipate the heat generated from the power transistor.

5.8.3 Heat Sink

A metal plate specially designed to conduct and radiate heat from an electronic component. A layer of material place within the outer skin of high-speed aircraft to absorb heat.

5.9 TRANSISTOR CONFIGURATIONS

A transistor is a three terminal device, (i.e., base, emitter, collector). But it require four terminals for connecting it in circuits.

(i.e.) Two terminals for input, two terminals for output.

Hence one of the terminal is made common to the input and output circuits. Common terminal is grounded.

Types of Configuration

Three types of configuration is available

1. *Common Base configuration (CB)*
2. *Common Emitter configuration(CE)*
3. *Common Collector configuration(CC)*

5.9.1. Common Base Configuration

Common base configuration circuit is shown in figure 5.13. Here base is grounded and it is used as the common terminal for both input and output.

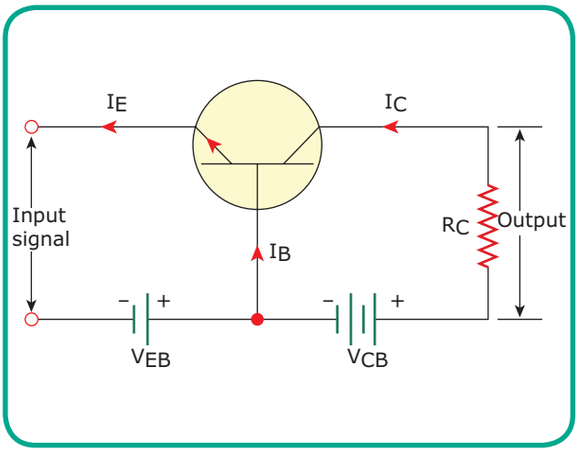


Figure 5.13 Common base configuration

It is also called as grounded base configuration. Emitter is used as an input terminal whereas collector is the output terminal.

Input characteristics:

It is defined as the characteristic curve drawn between input voltage to input current whereas output voltage is constant.

To determine input characteristics, the collector base voltage V_{CB} is kept constant at zero and emitter current I_E is increased from zero by increasing V_{EB}. This is repeated for higher fixed values of V_{CB}.

A curve is drawn between emitter current I_E and emitter base voltage (V_{EB}) at constant collector base voltage (V_{CB}) is shown in figure 5.14 .When V_{CB} is zero emitter base junction is forward biased. So it behaves as a diode so that emitter current increases rapidly.

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